

PEQG 2026 — Poster-Side Cheat Card

One page. Glance, don't read. Keep at the poster.

The thesis in one breath

Pleiotropy (broad expression + connectivity) predicts **constraint**. Tissue specificity predicts **where adaptation acts** — African selection on broadly-expressed genes, Melanesian on tissue-restricted ones. LoF constraint doesn't predict the adaptive target ($p = 0.97$). **Pleiotropy constrains the type of variation that can fix, not which genes can adapt.**

The three figures

Fig	Shows	Headline stat
1 Tissue-specificity gradient	129 genes, τ -split heatmap; specific enzymes (TYR/TYRP1/DCT) on top, broad regulators (MITF/KIT/EDNRB) below	descriptive — sets the stage
2 Pleiotropy $\rightarrow$ constraint	Joint OLS, breadth + KEGG predict LOEUF, both independent	$\beta_{\text{breadth}} = -0.135$ ($p=.001$); $\beta_{\text{KEGG}} = -0.131$ ($p=.002$); VIF = 1.16
3 $\tau \rightarrow$ population-specific PBS	Within-network percentile, 4 branches	AFR $\tau -6.89$ ($p=.011$); MEL $\tau +5.28$ ($p=.048$); opposite signs ; EUR/SAS null

Numbers that anchor you

- **Gene set:** 129 (Raghunath 2015); $n = 127$ in LOEUF regression; 124 regions in PBS.
- **Populations:** AFR 747 · MEL **47** (30 HGDP + 17 SGDP) · SAS 790 · EUR 788.
- **MC1R paradox:** LOEUF = 1.97 (most LoF-tolerant) yet African missense selection — LoF metrics miss it.
- **Supporting stats:** Fisher OR = 0.10, $p = 0.046$ · τ contrast $p = 0.032$ · LOEUF non-difference $p = 0.97$.
- **Genome-wide percentiles (Melanesian):** KITLG 98.6 · ATF2 98.6 · HIF1A 97.7 · PRKACA 96.9.

One-line “why this method” answers

- **LOEUF** — field-standard continuous LoF-constraint; checked ancestry-stratified, concordant.
- **τ** — standard tissue-specificity index, level-independent; report breadth too for robustness.
- **Hudson FST** — *unbiased for unequal/small n* (matters at $n=47$ MEL).
- **PBS** — isolates the target population's own branch from 3 pairwise FSTs.
- **Z-scoring** — puts predictors on one scale so β 's are comparable.
- **VIF 1.16** — breadth & KEGG essentially independent $\rightarrow$ two real axes of pleiotropy.
- **Added-variable plots** — slope = partial coefficient $\rightarrow$ visual proof of independence.
- **Percentile + permutation** — PBS is skewed & bounded; ranking is outlier-robust, permutation is assumption-free.

- **Spearman (rank-invariant)** — proves the SGDP sign-flip is a *dataset* effect, not the percentile transform.
-

Disclose these proactively (it reads as careful, not weak)

1. **“Consistent with selection,” never “evidence of.”** PBS = differentiation; drift can mimic it.
 2. **Within-network distribution, NOT network-vs-genome enrichment** (that test is null — and that’s fine, never claimed it).
 3. **Combined dataset for Fig 3**, not SGDP-only (SGDP genome-wide is null — sample size, $n \approx 18$ AFR).
 4. **$n = 47$ Melanesians** — biggest limitation; widens CIs, doesn’t bias point estimates.
-

If your mind goes blank — fallback phrases

- “Great question — I haven’t tested that directly; my instinct is ___, but I’d check first.”
- “That’s on the list for the publication version.”
- “Say more about what you’d expect to see?” (turns it into a conversation)
- “Let me write that down.” (*then do it*)

“I don’t know,” said calmly, is scientific maturity — not a miss. Bluffing is the only real mistake.

You are the world expert on this specific analysis. Nobody in the room has spent more time on these genes, these populations, these tests than you.